

# MEGHANA PISE

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## EDUCATION

**Dayananda Sagar College of Engineering, Bengaluru** | 2023 – 2027

B.E. in Computer Science Engineering (IoT, Cybersecurity & Blockchain Technology) | **CGPA: 8.95 / 10**

**Relevant Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning Basics, System Design Fundamentals

## TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript, C, SQL, HTML, CSS

**AI & Machine Learning:** Machine Learning fundamentals, LLM fundamentals, Prompt Engineering, LightGBM, PyTorch, FastAPI for AI applications, Streamlit, Ollama (running local open-source LLMs)

**Currently Building Proficiency In:** Generative AI, Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, end-to-end AI application development

**IoT:** ESP32, Arduino, Sensor Integration, GPS Integration, Firebase, IoT System Design

**Databases:** MySQL, Firebase Firestore

**Tools & Platforms:** Git, GitHub, VS Code, Postman, Figma

## PROJECTS

**EDNA — Environmental DNA Analysis System** [Code](#)

**Technologies Used:** Python, FastAPI, React (Vite), LightGBM, NCBI Entrez API

- Collaborated on a full-stack ML pipeline for eDNA-based species identification, contributing across data processing, model integration, and the React dashboard.
- Took primary ownership of training the LightGBM classification model using ~5,000 curated eDNA sequences sourced from the NCBI Entrez API, enabling top-3 species predictions with confidence scores.
- Contributed to the FastAPI backend and feature-extraction pipeline handling large, noisy biological datasets.
- Supported the feedback loop that enables periodic model retraining as verified predictions are added back into the training set.

**DART — Dynamic Routing & Alert Triage (SOC Automation)** [Code](#)

**Technologies Used:** Node.js, Next.js, Docker, Server-Sent Events (SSE)

- Collaborated on a SOC automation platform that enriches security alerts in real time using threat-intelligence APIs (GreyNoise, AbuseIPDB, VirusTotal, GeoIP) with automated risk scoring.
- Took primary ownership of the automated playbook system, implementing the decision logic that triggers mitigation actions (IP blocking, rate-limiting, service restart) based on computed risk scores.
- Contributed across the stack, including real-time alert delivery via Server-Sent Events (SSE) and the Docker-based service architecture.
- Validated system behavior under load using a simulated DDoS scenario (50 concurrent requests every 500ms for 60 seconds), with risk scores scaling to 40–95 as expected.

**Bone Cancer Detection & Classification (X-Ray Imaging)** [Code](#)

**Technologies Used:** PyTorch, ResNet-50, Owl Search Algorithm, Grad-CAM, Streamlit

- Built an end-to-end deep learning pipeline to classify X-ray images into normal, benign, and malignant categories using a ResNet-50 backbone with a custom classification head.
- Applied a bio-inspired optimization algorithm (Owl Search Algorithm) for automated hyperparameter tuning — exploring new parameter combinations while refining around those yielding better validation performance.
- Implemented Grad-CAM visualization to generate heatmaps highlighting the image regions driving each prediction, improving model interpretability.
- Built a Streamlit dashboard for real-time inference and prediction history, with simulated Big Data logging (Cassandra + HDFS-style storage) for production-style traceability.

**Bias-Free Text Checker** [Code](#)

**Technologies Used:** React, Tailwind CSS, JavaScript, FastAPI (Python), Speech Recognition & Synthesis APIs, Recharts

- Co-developed a tool that detects biased or non-inclusive language (including specific corrections like swapping "chairman" for "chairperson" and "guys" for "everyone") and suggests neutral alternatives in real time.
- Implemented regex-based text analysis on a FastAPI backend, paired with a React frontend supporting both typed and spoken input via the Web Speech API (speech-to-text and text-to-speech).
- Built an analytics dashboard using Recharts to surface detection trends, with analysis records logged locally for reference.

## CERTIFICATIONS & ACHIEVEMENTS

- Co-contributor on an IoT-Based Cold Chain Traceability System — currently under patent filing.
- BE10X AI Automation Workshop — hands-on exposure to modern AI workflows and tooling.
- ESP32 Workshop — practical experience with IoT and embedded systems development.

## HACKATHONS

- Sandbox 24-Hour Hackathon — Collaborated under strict time constraints to design and build a real-time alert routing system, focusing on rapid prototyping and system integration.
- Smart India Hackathon — Worked on eDNA-based biodiversity analysis, contributing to data processing, model design, and solution development in a competitive multi-team setting.